University of Bahrain

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TEMPERATURE MONITORIN SYSTEM

Mini-Project – PICENG321

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ABSTARCT

This report is about a mini project that points out how to build a temperature monitoring system to control a patient's temperature continuously using a temperature sensor. The temperature sensor is connected to an ADC of the microcontroller. Then the microcontroller is connected to a WIFI that can transmit the patient temperature to the nursing staff in a hospital wirelessly. Also, the microcontroller is connected to an LCD which displays the patient temperature all the time. Each time a patient's temperature reaches higher than 38 degrees Celsius, the buzzer will ring for notification. Lastly, all reading of patient's temperature will be saved on an online platform so the hospital staff can access it anytime as a reference.

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1.0 Objectives

- To understand how to program a microcontroller to perform certain instructions provided to create a temperature monitoring system and program it practically.
- Master how to read temperature readings from a temperature sensor.
- Accomplish a project that continuously controls and monitors a patient's temperature.

2.0 INTRODUCTION

From smallpox to measles, viral hemorrhagic fevers, and intense acute breathing infections, the human race has survived many infectious pandemics for the duration of the years. Todays pandemic that has affected all societies is the very well-known, Covid 19. The not unusual place factor among these kind of sicknesses is that each one the ones pandemics have been contagious. Focusing on Coivid-19, a number of the signs and symptoms of this disorder are experiencing an growth withinside the human body. So detecting the temperature can paintings as a signal to signify if the individual can be inflamed or not. So this paper will take vicinity to speak approximately constructing a temperature tracking device to be utilized by scientific body of workers to soak up the temperature of the affected person in any vicinity below any conditions.

Temperature tracking structures seize temperature facts through a sensor and an alarm that permits you to understand whilst the temperature has risen and if the affected person is below scientific concern. A temperature alarm tracking device may be very beneficial as a warmth tracking device as it permits you to manipulate the temperature tiers withinside the human body. And the non-stop show of temperature studying on a screen, like LCD, can assist in understanding the sufferer's situation on the moment. The device may be managed via using a microcontroller and can be constructed via way of means of the use of more than one digital devices. A temperature tracking device generation will assist hold temperature efficacy and automate your compliance reporting via WIFI that may be without delay related to the health facility to show the affected person effects and take instantaneously actions.

3.0 WORKING PRINCIPLE

The working principle is simple. The temperature sensor, the one used in the project is the KY-013 analog temperature sensor, senses, and measures the temperature of the patients. The

microcontroller, NodeMCU, receives the temperature data as input from the sensor. Then, as the microcontroller gains the readings from the sensor, these readings will be presented or displayed constantly on the LCD (liquid crystal display). The microcontroller connects to a buzzer output. It will work by making a sound under one condition, and that's when the temperature reading is more than 38◦c. Lastly, to make a saved record of the temperature readings, the Wi-Fi module, esp8266 Wi-Fi module that's built in the NodeMCU, was used. The readings are transported to a secure platform that can be accessed by logging in into the account, which was specially made for this purpose on an online platform called MyDevices Cayenne.

4.0 PROGRAM ALGORTHM

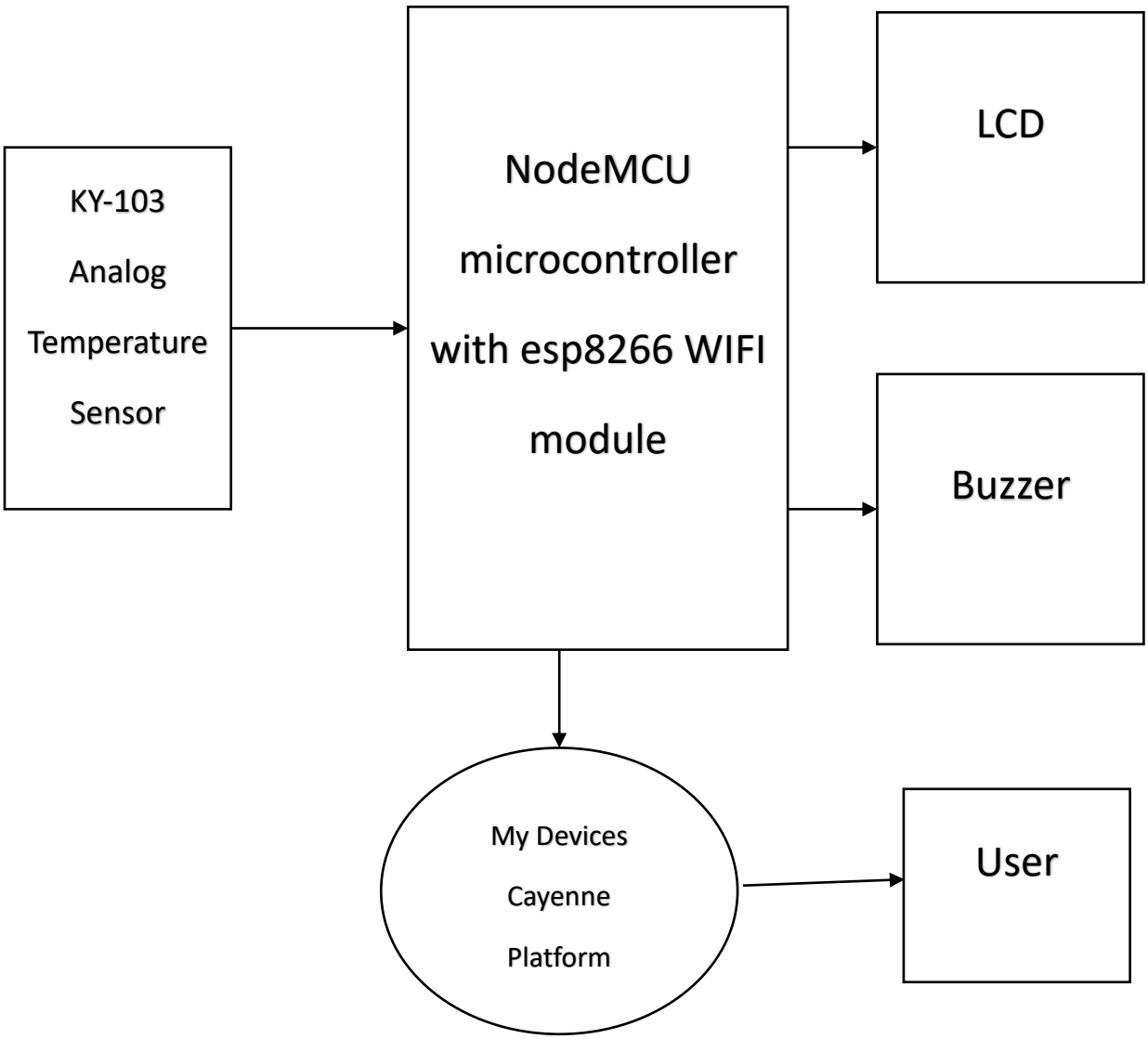


Figure 1. Block Diagram

5.0 COMPONENTS

1. KY-013 Analog Temperature Sensor

The analog temperature sensor actually acts as a variable resistor. The sensor reduces its voltage output as the temperature rises. Once we have scaled the voltage output, we calibrate the sensor and convert the voltage output to temperature. Analog temperature sensor KY-013 The module contains an NTC thermistor and a 10 kΩ resistor. The resistance of the thermistor varies with the ambient temperature. [1]

Operating Voltage	5V
Temperature measurement range	-55°C to 125°C [-67°F to 257°F]
Measurement Accuracy	± 0.5°C

Table 1. Sensor Specifications



Figure 2. KY-013 Analog Temperature Sensor

2. NodeMCU ESP8266

NodeMCU is an open-source development board specially designed for IoT (Internet of Things) applications. The NodeMCU ESP8266 development board comes with an ESP-12E module that contains an ESP8266 chip with a 32-bit microprocessor. This microprocessor works with an adjustable clock frequency from 80 MHz to 160 MHz. NodeMCU has 128 KB of RAM and 4 MB of flash memory for storing data along with programs. Its high processing power with a built-in Wi-Fi module system makes it ideal for IoT projects. It can be powered on using a Micro USB connector and VIN pin. Plus, it supports UART, SPI, and I2C interfaces. [2]



Figure 3. NodeMCU ESP8266 .

Microcontroller: Tensilica 32-bit RISC CPU Xtensa LX106
Operating Voltage: 3.3V
Input Voltage: 7-12V
Digital I/O Pins (DIO): 16
Analog Input Pins (ADC): 1
UARTs: 1
SPIs: 1
I2Cs: 1
Flash Memory: 4 MB
SRAM: 64 KB
Clock Speed: 80 MHz
USB-TTL based on CP2102 is included onboard, Enabling Plug n Play
PCB Antenna
Small Sized module to fit smartly inside your IoT projects

Table 2. NodeMCU Specifications

3. LCD (Liquid Crystal Display)

An LCD (Liquid Crystal Display) screen includes a thin layer of liquid crystal material sandwiched between two electrodes on glass substrates, with two polarizers on each side. A polarizer is an optical filter that lets light waves of a specific polarization pass through while blocking light waves of other polarizations. The electrodes need to be transparent, so the most popular material is ITO (Indium Tin Oxide). As LCD can't emit light itself, normally a backlight is placed behind an LCD screen to be seen during the dark environment. The light sources for the backlight can be LED (Light Emitting Diode) or CCFL (Cold Cathode Fluorescent Lamps). The color filter consists of RGB color. The principle behind the LCDs is that when an electrical field is not applied to the liquid crystal molecules, the molecules twist 90 degrees in the LCD cell. When the light either from ambient light or from the backlight passes through the first polarizer, the light is polarized and twisted with the liquid crystal molecular layer. When it reaches the second polarizer, it is blocked. The viewer sees the display is black. When an electric field is applied to the liquid crystal molecules, they are untwisted. When the polarized light reaches the layer of

liquid crystal molecules, the light passes straight through without being twisted. When it reaches the second polarizer, it will also pass through, the viewer sees the display is bright. Because LCD technology uses electric fields instead of electric current (electron passes through), it has low power consumption. [3]

Operating Voltage is 4.7V to 5.3V
Current consumption is 1mA without backlight
Alphanumeric LCD display module, meaning can display alphabets and numbers
Consists of two rows and each row can print 16 characters.
Each character is build by a 5×8 pixel box
Can work on both 8-bit and 4-bit mode
It can also display any custom generated characters
Available in Green and Blue Backlight

Table 3. LCD Specifications

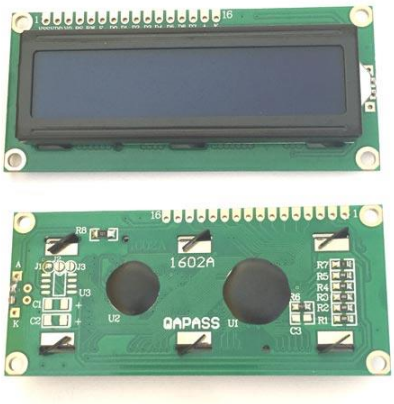


Figure 4. LCD

4. Buzzer

Buzzers are essentially tiny audio systems that could be related without delay to Arduino. It could make it sound a tone at a predefined frequency. The buzzer produces sound primarily based totally on the opposite of the piezoelectric impact that is an impact in which crystals will form whilst the following of power to them. By making use of an electric-powered sign on the proper frequency, the crystal could make a sound. The buzzer produces an equal noisy sound no matter the voltage variant implemented to it. It includes piezo crystals among conductors. When an ability is implemented throughout those crystals, they push on one conductor and pull at the other. This, push and pull action, outcomes in a valid wave. Most buzzers produce sound withinside the variety of two to four kHz. An Arduino buzzer is likewise known as a piezo buzzer. [4]

Rated Voltage: 6V DC
Operating Voltage: 4-8V DC
Rated current: <30mA
Sound Type: Continuous Beep
Resonant Frequency: ~2300 Hz
Small and neat sealed package
Breadboard and Perf board friendly

Table 4. Buzzer Specifications



Figure 5. Buzzer

5. Jumper Wires

Jumper wires are wires that have connector pins at every end, letting them be used to attaching factors to every different without soldering. Jumper wires are commonly used with breadboards and different prototyping equipment to make it smooth to alternate a circuit as needed. Fairly simple. Though jumper wires are available in quite a few colors, the colors do now no longer surely imply anything. This manner that a crimson jumper cord is technically similar to a black one. But the colors may be used for your gain to differentiate among varieties of connections, inclusive of floor or power. Jumper wires are commonly available in 3 versions: male-to-male, male-to-lady, and lady-to-lady. The distinction among every is in the long run factor of the cord. Male ends have a pin sticking out and might plug into matters, even as lady ends do now no longer and are used to plug matters into. [5]



Figure 6. Jumper Wires

6. Breadboard

A breadboard is a solderless tool for the transient prototype with electronics and takes a look at circuit designs. Most digital additives in digital circuits may be interconnected via way of means of placing their leads or terminals into the holes, after making connections via wires wherein appropriate. The breadboard has strips of metallic below the board and joins the holes at the pinnacle of the board. The metallic strips are laid out as proven below. Note that the pinnacle and backside rows of holes are related horizontally and break up withinside the center whilst the closing holes are related vertically. There are forms of breadboards; those are solder and solderless forums. Solder forums are forums you need to solder additives onto (in line with the name). These

are the maximum of your trendy circuit forums, and in case you turn one over, you'll observe that every one of the connections is soldered to the board itself. [6]



Figure 7. BreadBoard

6.0 SOFTWARE

In this project we used ARDUINO IDE and mydevices Cayenne website for IoT.

1. Arduino IDE

Arduino IDE is an open-source software that is mainly used for writing and compiling the code into the Arduino Module. It makeس code compiling too easy that even a common person can understand and get used to it. It is available for operating systems like MAC, Windows, Linux. This application supports both C and C++ languages. It can be used for all Arduino modules and other also. This IDE has a serial monitor which can be used to display many things, which is important to check the working of the systems. The serial monitor is used to preview the serial data as written lines, while the serial plotter previews the data as a graph. Both are useful for different applications. [7]

Figure 8. Arduino IDE

2. MyDevices Cayenne Website

My Devices Cayenne permits you to speedy design, prototype, and visualize IoT solutions. You can use Cayenne as a device to visualize real-time and ancient data, despatched over The Things Network [8]. It is an answer for building IoT programs primarily based totally on structures referred to as Arduino, Raspberry pi, and ESP8266. Cayenne permits to ship values from and to

the dashboard, it additionally permits to preview the values in lots of methods consisting of tables and graphs [9].



Figure 9. My Devices Cayennce Website

6.1 Code

```
#include <ESP8266WiFi.h>

#include <WiFiClient.h>

#include <CayenneMQTTESP8266.h>

#include<Wire.h>

#include <hd44780.h>                       // main hd44780 header

#include <hd44780ioClass/hd44780_I2Cexp.h>

#ifndef STASSID

#define STASSID "XL 5000G"

#define STAPSK  "123467890"

const char* username = "7a4c8090-b85f-11eb-883c-
638d8ce4c23d";

const char* MQTTp =
"0577644c05a22660ab46bc27d72917d8cae68dd7";

const char* Client_ID = "826a98c0-b85f-11eb-8779-7d56e82df461";

#endif

hd44780_I2Cexp lcd;  // I2C address 0x27, 16 column and 2 rows

int temperaturePin = A0;

int buzzer = D4;
```

```
double tempsensor(int ADCreading)

{

double Temp;

Temp = log(10000.0*((1024.0/ADCreading-
1)));

Temp = 1 / (0.001129148 + (0.000234125 +
(0.0000000876741 * Temp * Temp ))* Temp
);

Temp = Temp - 273.15;            // Convert
Kelvin to Celcius

return Temp;

}

void setup() {

Cayenne.begin(username, MQTTp,
Client_ID, ssid, password);

Wire.begin(D2, D1);

lcd.begin(16, 2);

lcd.backlight();

Serial.begin(9600);

pinMode(A0,INPUT);
```

```
Serial.begin(9600);
WiFi.mode(WIFI_STA);
WiFi.begin(ssid, password);
Serial.println("");

// Wait for connection
while (WiFi.status() != WL_CONNECTED) {
delay(500);
Serial.print(".");
}
Serial.println("");
Serial.print("Connected to ");
Serial.println(ssid);
Serial.print("IP address: ");
Serial.println(WiFi.localIP());
}
void loop() {
Cayenne.loop();


int data=analogRead(temperaturePin);
double temp =  tempsensor(data);
Serial.print("Temperature = ");
Serial.print(temp);
Serial.print("°C");
Serial.println();
lcd.setCursor(0,0);
lcd.print("Temp in C");
lcd.setCursor(0,1);
lcd.print(temp);
delay(1000);
lcd.clear();


if(temp>38)
{
digitalWrite (buzzer, HIGH);
}


else
{
digitalWrite (buzzer, LOW);
}
delay(500);
Cayenne.virtualWrite(1, temp, TYPE_TEMPERATURE);
Serial.print("Temperature: ");
Serial.print(temp);
Serial.println(" C");
}
```

7.0 PROJECT DEMO

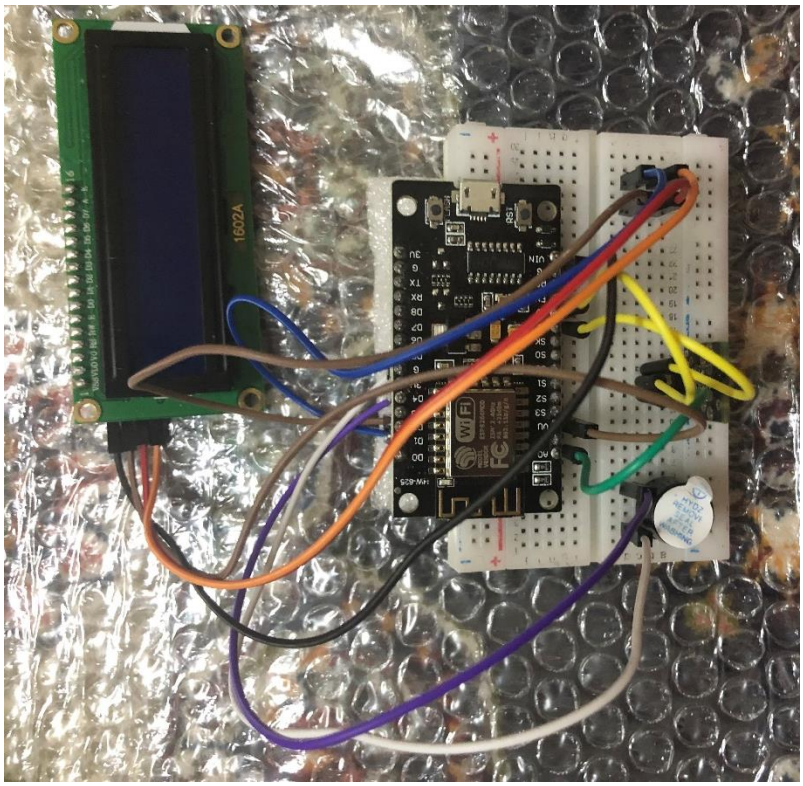


Figure 10. Connections

Please refer to the link to see the recorded videos that works as an evidence how the system worked.

https://drive.google.com/drive/u/0/folders/1Mvgp3xOUY23CWQC9XLjsnzyE6BfB6hAX

8.0 DIFFICULTIES

While preparing this project, some issues were faced. The most important one was when the Nodemcu kept on rejecting connection to the Wi-Fi. At first. It connected once, but then it kept on refusing to connect. Trying to solve the issue, another Nodemcu was bought and was checked. Yet, the same problem was faced. Therefore, the attention was then focused on the Wi-Fi network used to connect the Nodemcu to the internet. All other devices that were connected to the WIFI network were disconnected to reduce the interference between the Wi-Fi network itself and the Nodemcu. The problem was solved then.

Another point is that working on the project would have been much easier if it was done in the university or under different circumstances, where meetings and gatherings can be arranged. However, due to Covid-19, it was impossible.

9.0 CONCLUSION

To sum up, all the objectives were achieved successfully. The group was able to build up a system for monitoring patient's temperature, using different kinds of electrical technologies and instruments. A complete system that measures and displays temperature interminably was executed. Plus, the system notifies the user if the temperature went too high. All the data are sent through the internet to a secure online platform where the data is stored.

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